

Matin Horri

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PROFESSIONAL SUMMARY

Dynamic and results-driven Software Engineer experienced in developing and optimizing software for **medical devices**, web applications, and machine learning models. Skilled in **object-oriented** programming, with expertise across the full Software Development Life Cycle (SDLC), focusing on **developing, testing, analyzing**, and **documenting** software.

SKILLS

Programming Languages: C++, C, Python, C#, JavaScript, Java, SQL, Perl, Bash

Frameworks & Libraries: .NET, Google Test, Pandas, NumPy, Node.js

Tools: Linux shell (Bash) Scripting, Visual Studio, Git, Azure, Jenkins, Perforce, AWS, Docker, CI/CD

Debugging & Testing: GDB, Visual Studio Debuggers, Valgrind, Python-based Automation

Engineering Practices: Embedded Systems, Real-Time Applications, Agile Development, SDLC

Architectural Concepts: Multithreading (POSIX), RESTful APIs, SOLID Principles, Design Patterns

WORK EXPERIENCE

Abbott

Software Engineer-Medical Device

Minneapolis, Minnesota

April 2024- Present

- Contributed to testing and developing EnSite X Electrophysiology, a C++ real-time Medical Software on Linux, following embedded software development principles
- Collaborated on software automation, validation, and integration by executing Linux shell and Python scripts, and analyzing results from system-level stress testing and embedded network interface
- Participated in code reviews and debugging to identify root causes, resolve issues, and optimize software performance and stability, including memory and network usage, CPU efficiency, GPU, and latency
- Performed system-level testing and defect analysis with direct medical device interfacing, collaborating cross-functionally to define test strategies and ensure compliance with FDA, ISO 13485, and IEC 62304

Thomson Reuters

Software Developer Intern

Minneapolis, Minnesota

August 2022- January 2023

- Developed web features for the Practical Law platform using JavaScript, C#, and .NET for backend
- Used GitHub for version control, Visual Studio for development and debugging, and Microsoft Azure for CI/CD deployment
- Created and executed test cases to improve performance and reduce load times across the platform
- Utilized Visual Studio Debugger and Chrome DevTools to troubleshoot and optimize backend and frontend functionality

EDUCATION

University of Minnesota, Twin Cities (College of Science & Engineering)

December 2023

Bachelor of Science (HONORS)-Computer Science

Major GPA: 3.6

PROJECTS EXPERIENCE

Voting System | [LINK](#)

January 2023- May 2023

- Developed C++ voting system for closed party and instant runoff election, utilizing SDLC

- Created Software Design Document (**SDD**) and Source Document (**SRC**), and implemented unit and integration tests in Google **Test** format, achieving 90% test coverage

Autonomous Vehicle Transportation project (e.g., Uber++) | [LINK](#)

January 2022- May 2022

- Developed the Uber++ website, implementing data structure & **algorithms** (A*, DFS, Dijkstra) in **C++** for efficient pathfinding
- Integrated features like robotic customers and drones, performed unit and integration **testing**, and created comprehensive **documentation** and analysis **reports**
- Applied **design patterns** (Singleton, Factory) and adhered to **SOLID** principles utilizing **object-oriented** programming principles, improving code maintainability

Medical Note-Taking Android App | [LINK](#)

September 2022- December 2022

- Developed an Android note-taking app using **Java** and Android SDK, featuring note formatting, image insertion, hierarchical organization, and user login
- Conducted user research and usability testing to **enhance UX**, and utilized **RESTful API** integration, increasing user satisfaction

TODO List, Portfolio Website | [LINK](#)

September 2022- December 2022

- Developed a dynamic portfolio website using **Python**, **NodeJS**, **JavaScript**, **HTML** and **SQL**
- Built a **Python** server for **API** requests and utilized **MySQL** for data management

Snake Route AI | [LINK](#)

January 2023- May 2023

- Developed and implemented AI algorithms for route navigation in the Snake Game, exploring DFS, BFS, and A* search algorithms to improve pathfinding efficiency over 300 iterations
- Applied machine learning techniques to optimize route prediction, utilizing **Python**, **Pandas**, model training, supervised/unsupervised learning and **NumPy** to process data and visualize performance

Multi-Threaded Web Solutions | [LINK](#)

September 2023- December 2023

- Designed and implemented a **multi-process** web browser in **C** for stability through process isolation
- Developed a high-performance **multi-thread** web server managing concurrent HTTP requests using **POSIX** threads and optimizing **CPU usage**, increasing server throughput

Smart Hub Embedded Tablet – ACM Group Project

Dec 2022- Dec 2023

- Worked on the SmartHub Embedded Tablet System, an IoT-based embedded system that provides real-time transit updates, weather forecasts, and interactive gaming features
- Developed firmware in C for an Arduino microcontroller, and Optimized API communication

HONORS & CERTIFICATES

- Certified **Windchill** Advanced Engineering
- Member, Association for Computing Machinery (ACM)

Jan 2022 - Jan 2023